High-Frequency Trading in the Chinese Stock Market: Predictability, Profitability, and Regulation

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Preliminary Version

Abstract

We study the predictability of ultra-high-frequency stock returns and trade durations in the Chinese A-share market, extending the framework of Aït-Sahalia et al. (2022). The Chinese market—characterized by strict daily price limits, a dominant retail investor base, and distinct trading mechanisms—provides a compelling out-of-sample test for the universality of high-frequency predictability. Using comprehensive Level-2 order book data for a broad sample of stocks from 2021 to 2023, we employ machine learning methods, including LASSO and random forests, to forecast returns and durations over millisecond horizons.

Keywords: high-frequency returns, predictability, machine learning, chinese stock market, price limits, market microstructure.

JEL Codes: G12; G14; C45; C53; C58.

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1 Introduction

The predictability of asset returns is a central theme in financial economics. While long-horizon predictability remains a subject of debate (Fama, 1970; Lo and MacKinlay, 2002), recent evidence suggests that at ultra-high frequencies, market dynamics deviate significantly from a random walk. A seminal study by Aït-Sahalia et al. (2022), for instance, demonstrates that high-frequency stock returns and durations in the U.S. market are systematically and pervasively predictable over very short horizons, leveraging machine learning techniques to extract signals from vast datasets.

An open question, however, is whether these findings represent a universal feature of modern electronic markets or are contingent on the specific market structure of the United States. The U.S. market is mature, highly institutionalized, and operates without the strict constraints found in many other major economies. This paper addresses this question by replicating and extending the analysis of Aït-Sahalia et al. (2022) in the world's second-largest equity market: China.

The Chinese A-share market provides a unique and powerful laboratory for this investigation due to several key institutional differences:

- **Price Limits:** Stocks are subject to a strict $\pm 10\%$ daily price limit (price limit system). These hard boundaries fundamentally alter order flow dynamics, creating non-linearities and potential predictability patterns not observable in U.S. markets, especially as prices approach the upper or lower bounds.

- **Retail Investor Dominance:** Unlike the institutionally-dominated U.S. market, the Chinese market is characterized by a significantly higher proportion of trading volume from retail investors. This can lead to different patterns of noise trading, herding, and information processing, thereby creating distinct opporunities and risks for high-frequency traders (HFT).

- **Trading Mechanisms:** The T+1 trading rule and the constraints against short-selling impose significant intraday inventory constraints on market participants, potentially influencing short-term predictability.

These features motivate our primary research questions. First, does the strong high-frequency predictability documented in the U.S. persist in the structurally different Chinese market? Second, how do unique institutional features, particularly price limits, affect the magnitude and nature of this predictability? Third, are the same microstructure features (e.g., limit order book imbalance, transaction imbalance) the most important predictors, or does the different investor composition alter their relevance?

A growing literature documents that short-horizon predictability is not unique to the U.S. market. Using intraday Chinese equity data, Zhang et al. (2019) show that early-day returns (e.g., the first half-hour) significantly predict later intraday returns, while

Chu et al. (2019) find both momentum and reversal patterns within the trading day. Gao et al. (2019) further highlight that in China a substantial fraction of intraday return predictability is linked to overnight price movements, consistent with delayed information incorporation under the T+1 trading rule (Qiao and Dam, 2020). At the microstructure level, Shenoy and Zhang (2007) show that order imbalances predict future returns in Chinese stocks, in line with evidence from other markets that limit order book events and imbalance measures carry strong predictive content for short-horizon returns (Cont et al., 2014).

A second strand of literature emphasizes how the investor base in China shapes short-run return dynamics. Using detailed account-level data, Barber et al. (2008) show that retail trades can move prices, while more recent work by Tan et al. (2024) and Jones et al. (2025) documents that small retail investors in China tend to exhibit momentum-chasing and herding behavior, often generating return patterns that can be exploited by better-informed traders. This stands in contrast to the institutionally dominated U.S. market, where retail order flow is less central to overall price discovery (Boehmer et al., 2021). These findings suggest that the prominence of retail traders in China may amplify the predictive content of order-flow and volume-based signals at high frequencies.

Third, several institutional features of the Chinese market have been studied in isolation and are likely to affect high-frequency predictability. Daily price limits have been shown to generate both cooling-off and destabilizing effects (Wan et al., 2018; Chen et al., 2019), and the T+1 trading mechanism and short-selling frictions give rise to distinctive overnight and intraday return patterns (Qiao and Dam, 2020). In parallel, an emerging literature on high-frequency and algorithmic trading in China finds that high-frequency traders can be profitable, particularly in periods with strong microstructural frictions, although profitability appears to decline as markets mature and competition increases (e.g., Liu, 2021). Finally, there is a rapidly growing body of work applying modern machine learning techniques to return prediction in both U.S. and Chinese markets (Gu et al., 2020; Leippold et al., 2022), showing that nonlinear models can extract economically meaningful signals from rich sets of predictors.

Our paper brings these strands of literature together. We contribute to the high-frequency predictability literature by providing out-of-sample evidence from the Chinese A-share market; to the market microstructure literature by linking predictability directly to institutional features such as price limits and the T+1 rule; and to the emerging literature on high-frequency trading and machine learning in China by showing which limit order book and order-flow variables are most informative at millisecond horizons.

To answer these questions, we utilize high-resolution Level-2 data from the Shanghai and Shenzhen Stock Exchanges for all A-shares from 2019 to 2023. Following the methodology of Aït-Sahalia et al. (2022), we construct a rich set of predictors and employ a suite of machine learning models to forecast short-term returns. Our analysis provides robust

evidence on the generalizability of high-frequency predictability and offers new insights into its institutional determinants.

We find strong evidence of high-frequency predictability, confirming that this is a widespread phenomenon in modern electronic markets. For the median stock, we achieve an out-of-sample R^2 of 12.8% for 5-second return prediction and an accuracy of 64.8% for predicting the direction of the next trade. Our primary contribution lies in showing how this predictability is modulated by market structure. Specifically, our subsample analysis shows that predictability is significantly higher when a stock's price is close to its daily limit. Furthermore, our predictor importance analysis reveals that *statistics derived from transaction volume and order flow imbalance are even more dominant predictors in China compared to the United States. This finding potentially reflects retail herding behavior and the concentration of informed trading around specific price levels induced by the price limit mechanism.*

Our paper contributes to the literature in several ways. First, we provide crucial out-of-sample evidence on the generalizability of high-frequency predictability, demonstrating its robustness in a non-U.S. context. Second, we offer the first systematic evidence on how strict price limits impact predictability at the millisecond level, offering insights for market design and regulation. Finally, our findings have important implications for regulators in China and other emerging markets grappling with the rise of HFT.

The remainder of this paper is organized as follows.

2 Data

2.1 Data Description

We analyze comprehensive Level-2 order book data from both the Shanghai Stock Exchange (SSE) and Shenzhen Stock Exchange (SZSE), covering the period from February 23, 2021 to May 29, 2023. Our sample comprises 347 unique security symbols, representing the most actively traded securities in the Chinese A-share market. **Table 1** provides summary statistics for our dataset. Detailed descriptions of the Level-2 data structure and our complete data processing methodology are provided in Appendix A.2.

Our study focuses on the constituent stocks of the CSI 300 Index, which primarily consists of the largest and most liquid A-share firms and typically covers more than 70% of the total market capitalization of the Chinese equity market, playing a role comparable to that of the S&P 500 in the U.S. market. Because the index is rebalanced over time, this yields a stock universe of 347 distinct stocks that were included in the CSI 300 at least once during the sample period.

The Level-2 data report the top 10 price levels on both the bid and ask sides, together with the aggregate order size posted at each level, while transaction data record every

Table 1: Data Summary Statistics

Data Characteristic	Value
Securities included	All companies in CSI 300 on 2021–01–04
Number of different security symbols	347
Date range of data	2021–02–23 to 2023–05–29
Trading days included	506
Number of symbols available on all days	327
Number of available (symbol, date) pairs	172,340

executed trade with its timestamp, price, and size. This structure is similar to the NYSE Trade and Quote (TAQ) database, with the important difference that our quote data are richer as they contain depth information for multiple price levels beyond the best bid and ask. We exclude call-auction periods, covering only continuous trading hours (9:30–11:30 and 13:00–15:00), and merge the morning and afternoon sessions into a single intraday series for each stock and trading day.

2.2 High-Frequency Stock Returns

The prediction target in our study is the high-frequency stock return over four horizons, $\mathcal{H} = \{5, 15, 30, 60\}$ seconds. At each prediction time t, we forecast the Δ-second-ahead return using only information available strictly before t, with $\Delta \in \mathcal{H}$. To avoid ad hoc choices of evaluation times, we place a prediction point at every integer second throughout the continuous trading session (four hours per day in the Chinese stock market), yielding 14,400 ($= 3{,}600 \times 4$) prediction points per stock per day. Let $\mathcal{T}$ denote this set of prediction times.

The key to computing high-frequency returns is to obtain a well-defined transaction price at each prediction time. We use the transaction tape to assign a price to any $t \in \mathcal{T}$ during trading hours. Specifically, let $\mathcal{T}_i$ be the set of all trade timestamps for stock i. For any $t \in \mathcal{T}$, find the most recent trade time strictly before t, denoted $t_i^{\text{TRD}} \in \mathcal{T}_i$, and set the price for stock i at time t to the corresponding last-trade price $P_{i,t} = P_{i,t_i^{\text{TRD}}}$.[1] Given a horizon $\Delta \in \mathcal{H}$ and any $t \in \mathcal{T}$, the Δ-look-ahead return over $(t, t+\Delta]$ is

$$R_{i,(t,t+\Delta]} = \frac{P_{i,t+\Delta}}{P_{i,t}} - 1. \tag{1}$$

The return measure in (1) uses completed transactions at the relevant endpoints and differs from Aït-Sahalia et al. (2022), which relies on the average of transaction prices over the Δ window. We adopt this construction for three reasons. First, nearest-timestamp matching does not require a trade or quote exactly at t, so returns can be evaluated

[1] For example, if $t =$ 9:35:01.056745628 and the most recent trade occurred at 9:35:00.621280985 with price 6.88 RMB, then $P_{i,t} = 6.88$ RMB.

at any second, consistent with investors who want forecasts at arbitrary decision times. Second, many strategies depend on the price at the end of the horizon, which makes the endpoint to endpoint definition operationally natural (see Section X). Third, when liquidity is ample, as is typical in the Chinese market, the two measures are very close: the endpoint-based returns are highly correlated with the within-window averages, and the differences are small in practice. As shown in Online Appendix X, the gap between our measure and that of Aït-Sahalia et al. (2022) is modest and does not materially increase variance.

2.3 Return Predictors

To predict high-frequency stock returns, we construct a comprehensive set of predictor variables that characterize the short-term trading environment. Our predictor set includes 18 unique signals. We first follow Aït-Sahalia et al. (2022) by including 13 predictors that have been shown to possess predictive power in the U.S. equity market. We then augment this set with five additional predictors that are tailored to the unique characteristics of the Chinese stock market, including daily price restrictions, the high proportion of retail investors, and distinctive intraday trading patterns documented in prior research and industry practice.

Similar to Aït-Sahalia et al. (2022), all predictors are derived exclusively from time-stamped transaction and quote data recorded before the prediction time T.

A critical design choice concerns the length of the lookback window. We employ multiple disjoint lookback intervals spanning from very recent activity (seconds scale) to more distant history (minutes scale), allowing the machine learning algorithm to identify the most informative time horizons for each predictor without prejudging the results. Specifically, we define lookback spans (Δ_1, Δ_2) where $\Delta_1 = 0$ and $\Delta_2 \in \{3, 6, 12, 36, 72, 144, 216, 288\}$ seconds, creating eight distinct time windows for feature construction.

For timestamp T and lookback span (Δ_1, Δ_2), we define the lookback interval as:

$$\mathrm{Int}^{\mathrm{back}}(T, \Delta_1, \Delta_2) = \{t : T - \Delta_2 \leq t < T - \Delta_1\}. \tag{2}$$

Let $\mathcal{D}^{\mathrm{txn}}$ denote the set of all transaction timestamps, and let V_t, P_t, and Dir_t denote the volume, price, and direction (using the Lee and Ready (1991) algorithm) of the transaction at time t, respectively. For any time t, let P_t^a and P_t^b denote the best ask and bid prices, and S_t^a and S_t^b denote the aggregate order sizes at the best ask and bid levels.

We construct a comprehensive set of predictors grouped into five categories: (1) Volume and Trading Intensity, (2) Return and Price Dynamics, (3) Order Imbalance and Trading Direction, (4) Trading Cost and Spread, and (5) Order Book Depth Features. Detailed definitions and formulas for all factors are provided in Appendix A.3. Moreover, our feature engineering strategy employs a temporal expansion approach to capture the

dynamic evolution and distributional properties of market activity. Specifically, for a subset of base predictors exhibiting significant temporal variability, we compute the mean, standard deviation, and skewness across the eight lookback horizons. This approach, inspired by feature engineering methodologies from the Optiver Realized Volatility Prediction competition,[2] allows us to incorporate historical distributional information into the model.

Our final feature set comprises two distinct types: (1) *Base predictors with statistical transformations* (16 signals $\times$ 3 moments $\times$ 8 windows), and (2) *Direct predictors* (17 signals $\times$ 8 windows). This yields a total of 520 predictors available at each prediction time. Detailed specifications of the statistical transformations and the exact composition of the feature set are provided in Appendix A.4.

3 Methodology

3.1 Prediction Methods

We apply a set of widely used machine learning methods to predict high-frequency stock returns over the given prediction horizons. These methods fall into three categories: penalized linear regression models, tree-based models, and ensemble models. The penalized regression models include the least absolute shrinkage and selection operator (LASSO), ridge regression (Ridge), and elastic net (ENet). The tree-based models include random forest and light gradient boosting machine (LightGBM). The ensemble methods consist of a linear ensemble, defined as the simple average of the predictions from LASSO, Ridge, and ENet, and a tree ensemble, defined as the simple average of the predictions from random forest and LightGBM. In addition, we include an ordinary least squares linear regression model as a naive prediction benchmark.

These methods have been shown to perform well in a variety of financial prediction tasks, including the prediction of monthly stock returns, bond returns, and option returns across different markets. For the sake of brevity, we relegate detailed algorithmic descriptions and implementation choices to Section X of the Online Appendix.

3.2 Training, Validation, and Testing

Machine learning performance at high frequency is often driven as much by how the model is trained as by the model class itself. We therefore use a walk-forward cross-validation design that keeps the time ordering intact. We also estimate models stock by stock

[2]See https://www.kaggle.com/competitions/optiver-realized-volatility-prediction. We particularly reference the feature engineering approach of the first-place solution by nyanp: https://www.kaggle.com/code/nyanpn/1st-place-public-2nd-place-solution.

rather than pooling across securities. This lets predictability differ across names and avoids forcing the same coefficients or the same tuning choices on the entire cross-section.

A full grid search for every stock is computationally prohibitive, so we use a two-stage tuning procedure that concentrates the search where it matters. In the first stage, we randomly draw 10% of stocks and tune each model over a broad initial grid. For each sampled stock and each candidate configuration, we train on the most recent five trading days and evaluate on the next five trading days. We repeat this walk-forward exercise three times with rolling windows and track which configuration performs best. The configuration that wins most often provides an anchor for the second stage. We then build a compact refined grid by exploring a local neighborhood around the anchor values and apply this refined grid to the full universe. This approach greatly reduces computation while still allowing tuning to be done at the stock level.

We separate tuning from out-of-sample testing using nested rolling windows, as illustrated in Figure 1. The procedure runs in 40-trading-day cycles. The first 20 days are used only for hyperparameter selection, implemented through three consecutive 5-day training / 5-day validation rounds. In round $r \in \{1, 2, 3\}$, we train on days $T + 5(r - 1)$ through $T + 5r - 1$ and validate on days $T + 5r$ through $T + 5r + 4$, where T denotes the start of the cycle. This produces 15 daily validation R^2 values for each configuration. To keep tuning from being driven by a single unusual day, we drop the highest and lowest validation R^2 values and choose the configuration that maximizes the average of the remaining 13 observations.

Once hyperparameters are fixed, we evaluate performance out of sample over the next 20 trading days, $t = T + 20, \ldots, T + 39$. For each testing day t, we refit the model using only the most recent five trading days, $t - 5$ through $t - 1$, and generate a one-day-ahead forecast for day t. After the 20-day testing block, we roll the cycle forward by 20 trading days ($T \leftarrow T + 20$) and repeat the entire procedure, including re-tuning. This design keeps hyperparameter choice strictly on pre-test data and ensures that all reported results are genuinely out of sample, while the short daily re-estimation window allows the models to adjust to shifting high-frequency dynamics.

3.3 Evaluation Metrics

Our goal is to forecast high-frequency stock returns, so we evaluate each model along two standard and complementary dimensions: statistical fit and trading-relevant directionality. We use the out-of-sample R^2 to measure reductions in mean squared prediction error relative to a simple historical-mean benchmark, and directional accuracy (DA) to measure how often a model correctly predicts the sign of returns. The former is the natural metric for squared-loss prediction problems, while the latter is directly tied to sign-based trading rules and is less influenced by occasional extreme return realizations.

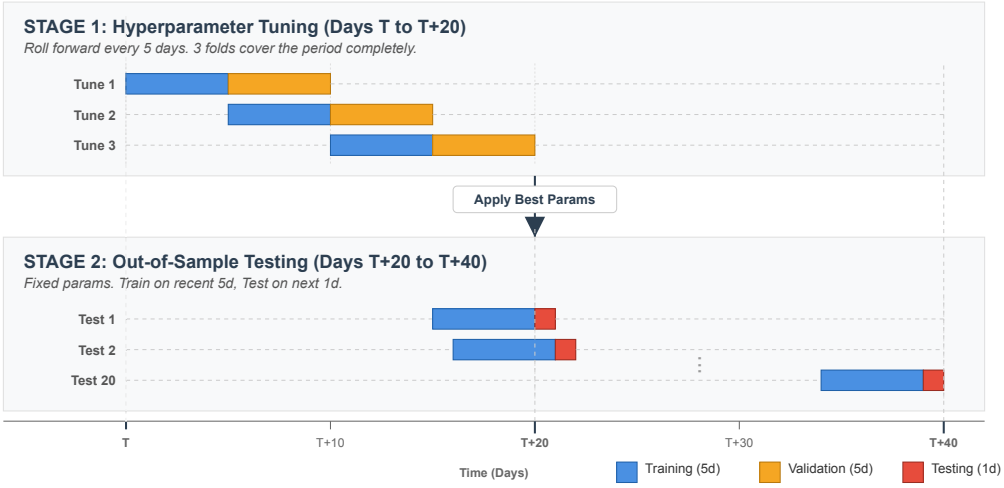


Figure 1: Two-Stage Walk-Forward Cross-Validation Strategy

Note: This figure illustrates the two-stage walk-forward cross-validation design. The procedure operates in 40-day cycles. Stage 1 (Hyperparameter Tuning) uses the first 20 days (T to $T+20$) to select the best model configuration via three rolling validation folds. Blue blocks represent 5-day training windows, and orange blocks represent 5-day validation windows. Stage 2 (Out-of-Sample Testing) covers the subsequent 20 days ($T+20$ to $T+40$). For each day t in this period, the model is retrained on the most recent 5 days (blue blocks) to forecast the next day (red blocks).

For stock i, let r_{it} be the realized return at prediction time t and let $\widehat{r}_{it}^{(m)}$ be the forecast from model m. Let $\mathcal{I}$ denote the set of stocks and $\mathcal{T}_i^{\text{oos}}$ the out-of-sample evaluation times for stock i. Let $\bar{r}_i$ be the unconditional mean return estimated from the corresponding training sample. We define

$$R^2_{\text{oos}}(m) = 1 - \frac{\sum_{i\in\mathcal{I}} \sum_{t\in\mathcal{T}_i^{\text{oos}}} \left(r_{it} - \widehat{r}_{it}^{(m)}\right)^2}{\sum_{i\in\mathcal{I}} \sum_{t\in\mathcal{T}_i^{\text{oos}}} \left(r_{it} - \bar{r}_i\right)^2}, \tag{3}$$

so that $R^2_{\text{oos}}(m) > 0$ indicates that model m improves on the historical-mean benchmark in terms of mean squared error. We also compute directional accuracy,

$$\text{DA}(m) = \frac{1}{N^{\text{oos}}} \sum_{i\in\mathcal{I}} \sum_{t\in\mathcal{T}_i^{\text{oos}}} \mathbb{I}\left\{\text{sign}\left(r_{it}\right) = \text{sign}\left(\widehat{r}_{it}^{(m)}\right)\right\}, \tag{4}$$

where $N^{\text{oos}} = \sum_{i\in\mathcal{I}} |\mathcal{T}_i^{\text{oos}}|$. DA is the share of out-of-sample observations for which the model correctly predicts whether the return is positive or negative. We report both metrics because they emphasize different aspects of forecast quality: R^2_{oos} rewards accurate magnitude forecasts but can be negative in noisy high-frequency settings, while DA is informative for direction-based strategies but ignores return magnitudes.

3.4 Predictor Importance Measures

Beyond forecast accuracy, we ask which predictors matter most. Because "importance" is defined differently across model classes (e.g., coefficients in linear models versus split gains in trees), raw importance scores are not comparable. We therefore work with ranks and aggregate them across the models in Section 3, which delivers a stable, model-agnostic ordering.

Within each model, we compute importance in the standard way. For penalized linear regressions (LASSO, Ridge, and Elastic Net), predictor j's importance is $|\widehat{\beta}_j|$; predictors are standardized, so magnitudes are comparable. For random forest, importance is the mean decrease in impurity due to splits on the predictor. For LightGBM, importance is the total split gain attributed to the predictor across boosting iterations. We then rank predictors within each model, with rank 1 assigned to the most important predictor, and combine these rankings using a Borda-count average. Let $\mathcal{M} = \{m_1, \ldots, m_K\}$ denote the set of models and $\mathcal{P} = \{p_1, \ldots, p_P\}$ the set of predictors. If Rank_{jk} is the rank of predictor p_j under model m_k, its average rank is

$$\overline{\text{Rank}}_j = \frac{1}{K} \sum_{k=1}^{K} \text{Rank}_{jk}. \tag{5}$$

To make the aggregated ranking easy to interpret, we anchor it with a baseline "noise" predictor. We add a synthetic predictor p_0 with zero importance by construction, rank it alongside the true predictors in each model, and denote its average rank by $\overline{\text{Rank}}_0$. Our final importance score, which we term the *excess importance score* (EIS), is defined as:

$$\text{EIS}_j = \overline{\text{Rank}}_j - \overline{\text{Rank}}_0. \tag{6}$$

A score of $\text{EIS}_j = 0$ means predictor p_j is no better than an uninformative variable, while $\text{EIS}_j > 0$ means it ranks above noise on average. Larger values indicate predictors that consistently show up near the top across models.

4 Predictability of High-Frequency Stock Returns

This section examines the magnitude of high-frequency return predictability in the Chinese A-share market and explores the economic mechanisms driving these patterns. The analysis first establishes the predictability relative to developed markets, demonstrates the necessity of nonlinear modeling, and finally dissects the sources of predictability, linking them to market microstructure features such as retail dominance, the T+1 trading rule, and price limits.

4.1 The Pervasiveness of High-Frequency Predictability

The analysis begins by establishing the magnitude of return predictability. Table 2 (Panel A) presents the out-of-sample R^2 across models and horizons. Evidence indicates that high-frequency predictability in the Chinese market is economically significant, substantially exceeding levels documented in mature markets.

Specifically, at the 5-second horizon, the Random Forest model achieves a median out-of-sample R^2 of approximately 15.5%, while penalized linear models (LASSO, Elastic Net) follow closely at approximately 14.8%. For comparison, Aït-Sahalia et al. (2022) report an R^2 of roughly 10% for U.S. equities using similar methods. This 50% relative outperformance suggests that the Chinese market, despite its high turnover, retains significant informational inefficiencies at ultra-high frequencies.

Figure 2 visualizes the distribution of R^2 values across the cross-section. A key observation is the rapid temporal decay: as the horizon extends from 5 seconds to 60 seconds, the predictive power declines significantly. For instance, the Random Forest R^2 drops by nearly half. Moreover, the boxplots reveal a critical distinction in model stability: while linear models exhibit wide dispersion with long negative tails—indicating catastrophic failure in certain stock-days—regularized and tree-based models show tight, consistent positive performance distributions. This decay pattern supports the hypothesis of fast-moving price discovery: while inefficiencies exist, arbitrageurs and high-frequency traders actively correct mispricing within the first minute.

Furthermore, Panel B of Table 2 shows that directional accuracy remains robust at approximately 60% even as R^2 declines. This implies that while the *magnitude* of returns becomes noisier over longer horizons, the *direction* of price adjustments remains highly predictable—a critical feature for execution algorithms.

4.2 Nonlinearity and Model Superiority

The next step examines the necessity of nonlinear modeling by comparing predictive accuracy across different specifications. Table 3 reports the Diebold-Mariano test statistics for pairwise model comparisons across four prediction horizons. Two ensemble benchmarks are included: a linear ensemble, which averages forecasts from three penalized linear models (LASSO, Ridge, Elastic Net), and a tree ensemble, which averages forecasts from Random Forest and LightGBM.

Three key patterns emerge. First, the unpenalized linear model significantly underperforms all other methods across all horizons, yielding negative out-of-sample R^2 values. This underperformance highlights the presence of high-dimensional noise in tick-level data, which destabilizes simple OLS estimation (Gu et al., 2020).

Second, tree-based models, particularly Random Forest, consistently outperform both individual penalized linear models and the linear ensemble. In Panel A (5-second horizon),

Table 2: Out-of-Sample Predictive Performance Across Models and Horizons

	Prediction horizon			
Model	5s	15s	30s	60s
Panel A: Return prediction (out-of-sample R^2)				
Linear	-0.190	-0.603	-1.047	-1.723
	(2.651)	(5.906)	(9.152)	(14.188)
LASSO	0.148	0.142	0.117	0.081
	(0.091)	(0.101)	(0.106)	(0.105)
Ridge	0.144	0.135	0.111	0.077
	(0.090)	(0.100)	(0.104)	(0.101)
Elastic Net	0.148	0.142	0.118	0.082
	(0.091)	(0.101)	(0.107)	(0.105)
LightGBM	0.142	0.139	0.112	0.072
	(0.091)	(0.100)	(0.106)	(0.104)
Random Forest	0.155	0.143	0.115	0.077
	(0.090)	(0.102)	(0.106)	(0.109)
Linear ensemble	0.130	0.125	0.104	0.073
	(0.085)	(0.096)	(0.100)	(0.097)
Tree ensemble	0.133	0.127	0.102	0.069
	(0.084)	(0.096)	(0.100)	(0.099)
Panel B: Direction prediction (accuracy)				
Linear	0.584	0.590	0.580	0.564
	(0.083)	(0.079)	(0.077)	(0.075)
LASSO	0.596	0.611	0.607	0.588
	(0.081)	(0.078)	(0.078)	(0.094)
Ridge	0.594	0.608	0.604	0.590
	(0.075)	(0.073)	(0.074)	(0.077)
Elastic Net	0.596	0.611	0.608	0.592
	(0.078)	(0.076)	(0.076)	(0.081)
LightGBM	0.595	0.609	0.602	0.583
	(0.082)	(0.078)	(0.078)	(0.083)
Random Forest	0.596	0.609	0.605	0.584
	(0.083)	(0.081)	(0.080)	(0.087)
Linear ensemble	0.586	0.603	0.602	0.591
	(0.073)	(0.073)	(0.073)	(0.076)
Tree ensemble	0.586	0.602	0.600	0.583
	(0.079)	(0.078)	(0.078)	(0.083)

Notes. The table reports out-of-sample predictive performance for high-frequency return forecasts. Panel A reports out-of-sample R^2 relative to a historical-mean benchmark. Panel B reports directional accuracy. Forecasts are generated stock by stock using the rolling training, validation, and testing design described in Section 3.2.

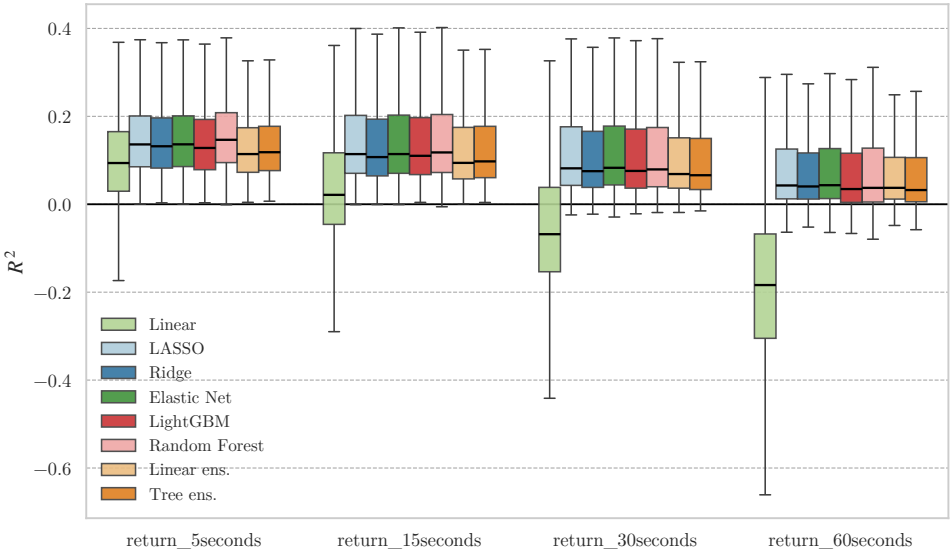


Figure 2: Comparative Performance of Machine Learning Models

Note: This figure displays boxplots of out-of-sample R^2 values across 5s, 15s, 30s, and 60s horizons. While linear models show high variance and negative medians at longer horizons, Random Forest (pink) and regularized linear models (LASSO, Elastic Net) demonstrate superior stability and consistently positive medians, identifying the robust signal in high-frequency data.

Random Forest significantly outperforms the linear ensemble ($t = 379.7$) and the tree ensemble ($t = 351.9$). This suggests that the bagging mechanism of Random Forest is particularly effective at handling the low signal-to-noise ratio in high-frequency data. The superiority of tree-based methods underscores the critical role of nonlinear interactions; for instance, the price impact of an order imbalance likely varies nonlinearly with the current bid-ask spread and the distance to price limits.

Third, the relative advantage of nonlinear models persists but evolves as the forecast horizon extends. While Random Forest maintains its lead, the magnitude of the Diebold-Mariano statistics against linear models decreases from the 5-second to the 60-second horizon. This suggests that while nonlinear microstructure effects are dominant at ultra-high frequencies, the signal decay allows linear components to capture a relatively larger share of the remaining predictability at the one-minute mark.

4.3 The Anatomy of Predictability

To understand the economic drivers of this predictability, the feature importance is dissected using the excess importance score. Figure 5 and Figure 3 decompose the contributions of different predictors, revealing a hierarchy of information that challenges standard microstructure assumptions.

First, and perhaps most strikingly, Figure 5 reveals that `PriceBias`—a metric capturing the proximity of the current price to the daily price limits—is the single most

Table 3: Pairwise Diebold–Mariano Tests for Return Forecasts

	Column model						
Row model	LASSO	Ridge	Elastic Net	LightGBM	Random Forest	Linear ens.	Tree ens.
Panel A: 5-second return forecasting (DM statistics)							
Linear	-51.34***	-50.77***	-51.4***	-50.38***	-52.43***	-48.78***	-49.3***
LASSO		194.68***	-36.92***	155.65***	-166.04***	334.0***	214.45***
Ridge			-209.53***	64.49***	-263.82***	270.14***	162.13***
Elastic Net				163.64***	-159.41***	338.9***	219.54***
LightGBM					-292.3***	205.27***	147.73***
Random Forest						379.7***	351.92***
Linear ens.							-120.5***
Panel B: 15-second return forecasting (DM statistics)							
Linear	-50.77***	-50.32***	-50.78***	-50.55***	-50.91***	-49.72***	-49.9***
LASSO		260.39***	-12.52***	89.6***	-39.87***	381.34***	235.84***
Ridge			-263.8***	-92.92***	-184.03***	223.92***	118.18***
Elastic Net				94.62***	-37.64***	385.35***	239.74***
LightGBM					-133.75***	270.66***	236.49***
Random Forest						304.46***	317.82***
Linear ens.							-88.51***
Panel C: 30-second return forecasting (DM statistics)							
Linear	-51.26***	-50.95***	-51.29***	-51.01***	-51.17***	-50.76***	-50.67***
LASSO		211.04***	-31.42***	142.03***	42.42***	284.83***	253.94***
Ridge			-223.52***	-29.87***	-111.7***	111.98***	128.71***
Elastic Net				154.97***	53.81***	291.48***	258.42***
LightGBM					-133.19***	115.76***	204.14***
Random Forest						180.31***	301.64***
Linear ens.							62.93***
Panel D: 60-second return forecasting (DM statistics)							
Linear	-51.07***	-50.95***	-51.08***	-50.84***	-50.95***	-50.9***	-50.77***
LASSO		98.21***	-22.66***	156.96***	69.58***	149.35***	178.27***
Ridge			-107.79***	77.54***	-0.55	45.47***	115.12***
Elastic Net				168.58***	76.55***	158.97***	187.18***
LightGBM					-92.41***	-38.27***	56.1***
Random Forest						28.47***	138.83***
Linear ens.							105.94***

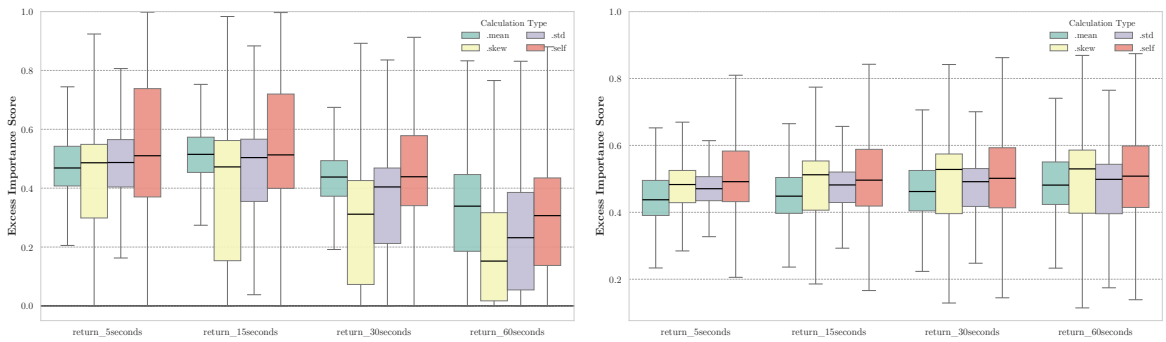
Notes. The table reports Diebold–Mariano test statistics for pairwise comparisons of out-of-sample R-squared performance across different forecast horizons. Panel A compares 5-second return forecasts, Panel B compares 15-second return forecasts, Panel C compares 30-second return forecasts, and Panel D compares 60-second return forecasts. A positive statistic indicates that the row model outperforms the column model and a negative statistic indicates the opposite. *** denotes significance at the 5% level. Linear ens. averages forecasts from the penalized linear models. Tree ens. averages forecasts from LightGBM and Random Forest.

powerful predictor across all horizons. This finding (darkest green in Figure 5) suggests that the "magnet effect" of price limits is the dominant force in Chinese high-frequency price formation. Unlike in U.S. markets, where LOB depth might dominate, in China, the structural constraint of the 10% limit creates a deterministic gravitational pull on prices that overrides standard order flow signals.

Second, the comparison of calculation types in Figure 3 highlights a divergence in how models extract information. The Random Forest model relies heavily on `.self` features (immediate, asset-specific values), achieving median importance scores near 0.7. In contrast, the Ridge regression places surprisingly high weight on `.skew` features (distributional asymmetry of order flow), particularly at longer horizons (30s and 60s). This implies that while linear models must aggregate distributional moments to find signal, nonlinear models can directly exploit the raw, immediate state of the asset.

Third, the temporal span analysis in Figure 4 uncovers a distinct "memory structure." For the 5-second prediction target, short-term lookback windows (3s and 6s) provide the most information, reflecting the fleeting nature of liquidity shocks. However, for the 60-second target, the importance shifts dramatically to the longest lookback window (288s). This aligns with the hypothesis that while arbitrage opportunities are transient, the broader momentum driven by retail herding requires a longer historical context to identify.

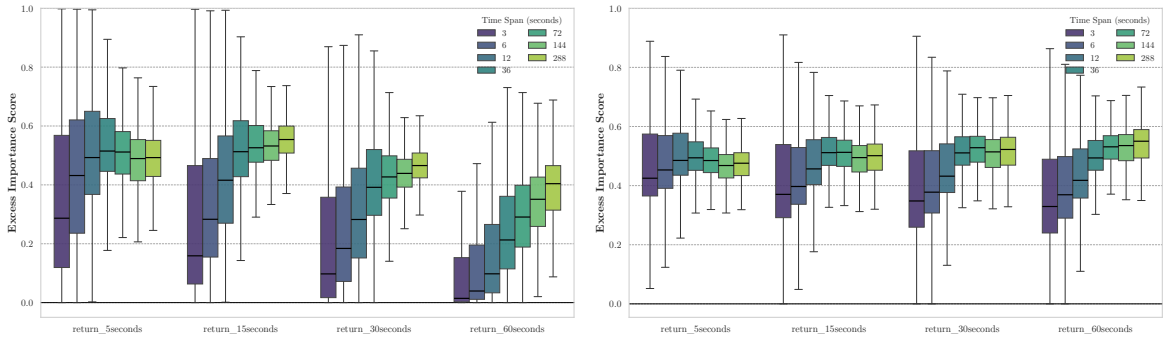
Figure 6 further details this by comparing the top 10 features. The Random Forest top list is dominated by `PriceBias` at various lags (e.g., `PriceBias_0_36`, `PriceBias_0_72`), confirming its nonlinear predictive power. Ridge, conversely, prioritizes linear aggregations like `VolumeSideSum.skew`, attempting to capture the asymmetry of buying and selling pressure through linear coefficients.



(a) Random Forest **(b)** Ridge Regression

Figure 3: Feature Importance Rankings: Random Forest vs. Ridge

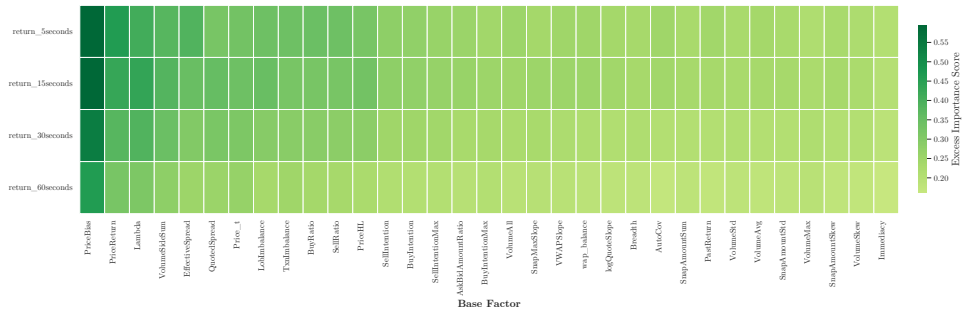
Note: Comparison of feature importance by calculation type. Random Forest (left) heavily utilizes '.self' (immediate asset state) features, while Ridge (right) increasingly relies on '.skew' (distributional asymmetry) features at longer horizons, suggesting linear models require higher-order moments to capture signal.



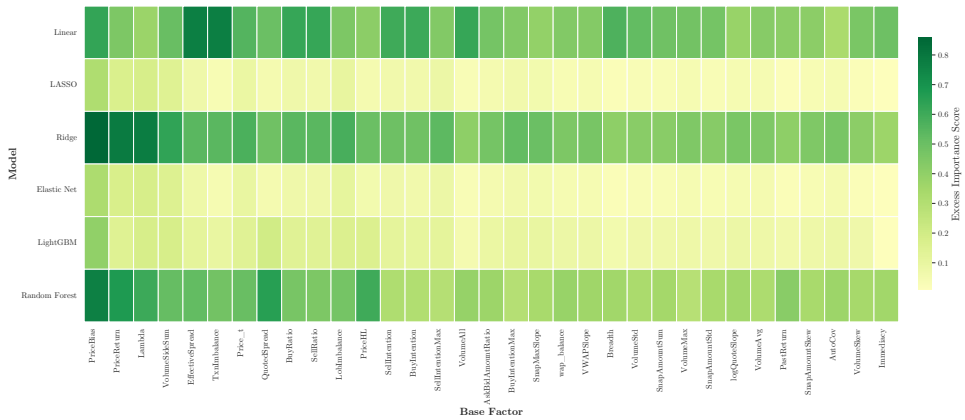
(a) Random Forest **(b)** Ridge Regression

Figure 4: Feature Importance by Time Span

Note: Feature importance decomposed by lookback window. For short-term targets (5s), short windows (3-6s) are critical. For longer targets (60s), information from the longest window (288s, yellow box) becomes dominant, indicating a shift from liquidity-based to momentum-based predictability.



(a) Feature Importance by Target



(b) Feature Importance by Model

Figure 5: Decomposition of Feature Importance

Note: Heatmap of excess importance scores. The darker green for 'PriceBias' across all targets (Panel a) and models (Panel b) identifies price limit proximity as the most robust predictor in the Chinese market, outperforming standard volume and spread metrics.

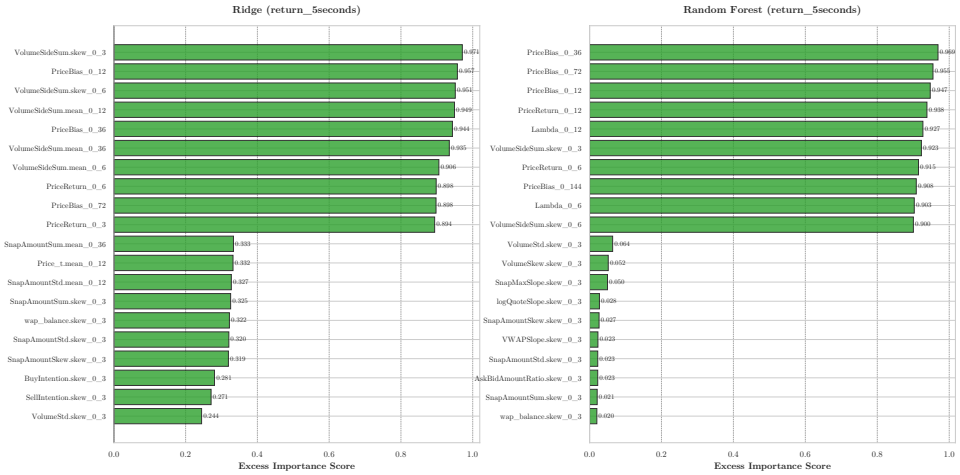


Figure 6: Top and Bottom Feature Analysis

Note: The figure displays, from top to bottom, the 10 most important and 10 least important features for model prediction. Random Forest exploits 'PriceBias' and 'Lambda' (price impact), while Ridge relies on linear 'VolumeSideSum' skewness, highlighting the difference in how models process market information.

4.4 Intraday Dynamics

High-frequency predictability is not uniform throughout the trading day. The intraday pattern, shown in Figure 7, reveals a distinct structure that maps to China's unique market design.

For the 5-second horizon, a distinct U-shaped pattern is observed in the morning session: predictability starts low, rises rapidly as the market digests opening news, and stabilizes. In the afternoon, it rises continuously, peaking near the close. This aligns with inventory management theories where traders rush to close positions before the end of the day. Interestingly, for the 60-second horizon, the pattern resembles a "W-shape," with a notable dip and recovery around the lunch break (11:30–13:00), suggesting that the midday pause introduces a "second opening" effect where accumulated information noise temporarily degrades longer-horizon predictability.

Additionally, the analysis considers the speed of information incorporation. Figure 8 presents the decay of the information coefficient (IC) as the signal is delayed. The results are striking: the half-life of the signal is approximately 1–2 seconds for the 5-second target and only 3–4 seconds for the 60-second target. This extremely rapid decay—much faster than the minute-level adjustments seen in some developed markets—underscores the ferocious competition in Chinese high-frequency trading. The window for arbitrage is not minutes, but mere seconds, mandating low-latency execution infrastructure.

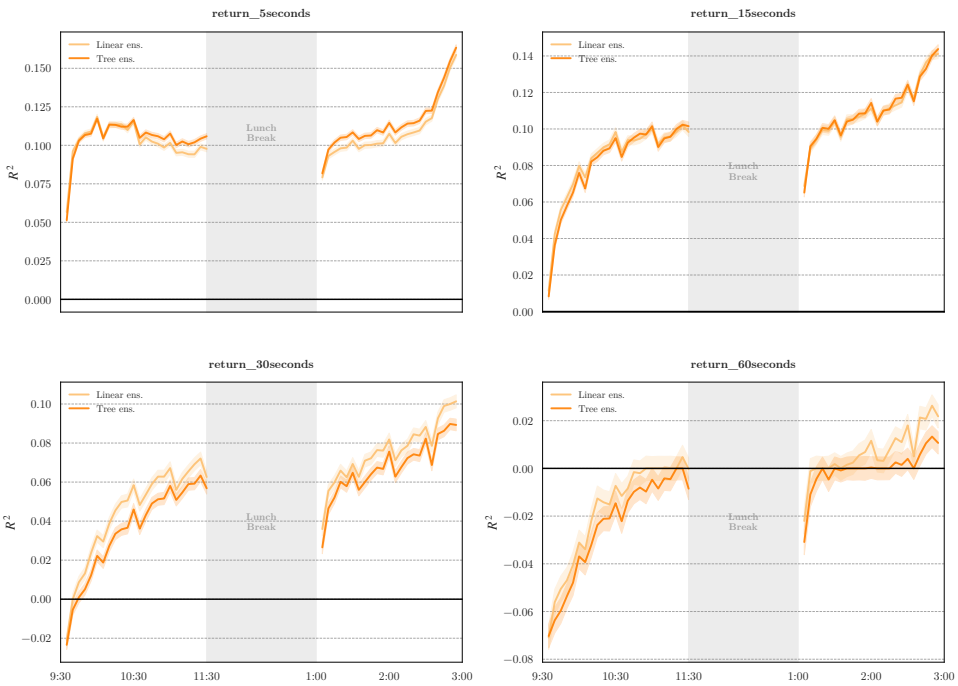


Figure 7: Intraday Variation in Predictive Performance: R^2 Analysis

Note: Intraday evolution of out-of-sample R^2. The curve is computed using a rolling window method, updated every 15 minutes based on data from the preceding 15-minute interval. The 5-second horizon (top left) shows a U-shaped morning and rising afternoon trend, while the 60-second horizon (bottom right) exhibits a W-shape with a post-lunch dip, reflecting the complex interplay of information flow and trading pauses.

4.5 Cross-Sectional Determinants of Predictability

To understand the economic mechanisms driving high-frequency predictability, the analysis examines how cross-sectional stock characteristics explain the variation in out-of-sample R^2 (see Appendix 13 for definitions and summary statistics of all features). Panel regressions are estimated in the form:

$$R^2_{i,t,h} = \alpha + \beta \mathbf{X}_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t} \tag{7}$$

where $R^2_{i,t,h}$ denotes the out-of-sample predictability for stock i on day t at horizon h, and $\mathbf{X}_{i,t}$ represents a vector of stock characteristics. Both stock (μ_i) and date (λ_t) fixed effects are included to absorb unobserved heterogeneity and time-varying market conditions.

4.5.1 Baseline Microstructure Frictions

Table 4 reports the results using standard microstructure controls. The findings establish a fundamental link between market frictions and predictability.

First, a strong Liquidity Friction Effect is documented. The coefficient on *Proportional Spread* is consistently positive and highly significant across all horizons (t-stat > 45).

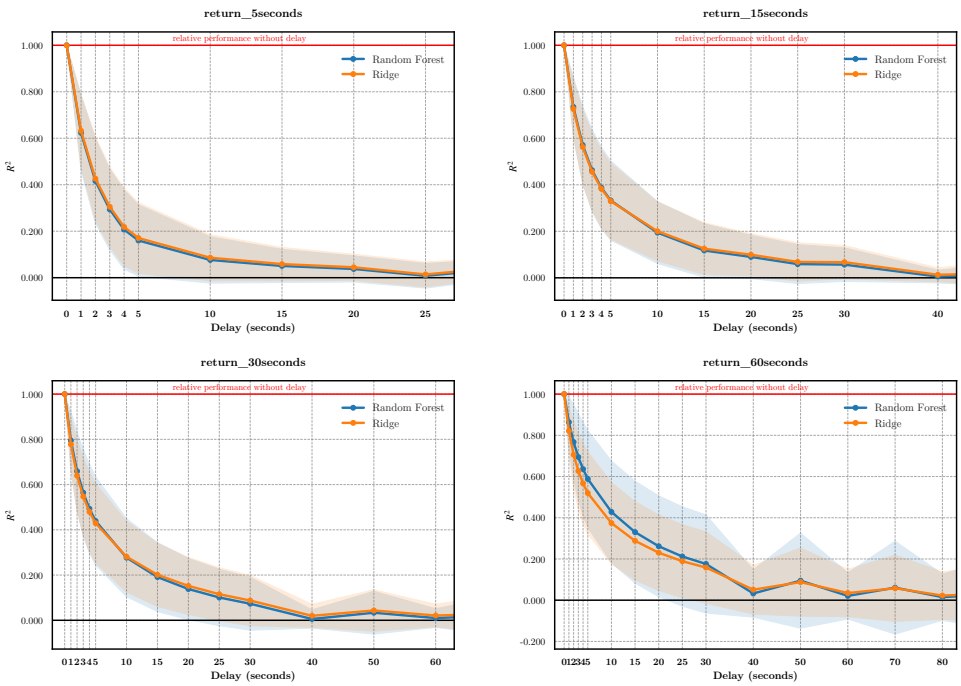


Figure 8: Signal Decay Performance: Correlation Analysis

Note: Decay of the Information Coefficient (IC) with signal delay. The half-life is estimated at 1–2 seconds for short horizons and 3–4 seconds for long horizons, indicating an ultra-efficient market at the micro-level where information is incorporated almost instantaneously.

This relationship is economically intuitive: high transaction costs (wide spreads) impose a "limit to arbitrage," preventing sophisticated traders from correcting small mispricings until they exceed the spread cost. Consequently, predictable price patterns accumulate in illiquid stocks.

Second, the *Volatility* coefficient is significantly negative. This serves as a control for signal quality. In periods of extreme turbulence, the signal-to-noise ratio deteriorates, making precise point forecasting more difficult for machine learning models.

Third, the *Traded Volume* exhibits an intriguing horizon-dependent pattern. It is positively associated with predictability at the 5-second horizon but becomes significantly negative at the 60-second horizon. This foreshadows the role of investor composition: high volume often correlates with retail activity (generating short-term noise), while at longer horizons, high liquidity facilitates efficiency.

4.5.2 Institutional Constraints and Investor Composition

Table 5 augments the baseline model with China-specific variables, revealing four distinct economic channels.

First, retail trading creates transient "noise alpha." The coefficient on *Retail Trade Ratio* is significantly positive at the 5-second horizon (0.016***) but flips to negative by 60 seconds (−0.007***). This reversal suggests that retail-driven momentum is short-lived

Table 4: Baseline Determinants of Predictability

VARIABLES	(1) Ridge 5s	(2) Ridge 15s	(3) Ridge 30s	(4) Ridge 60s	(5) RF 5s	(6) RF 15s	(7) RF 30s	(8) RF 60s
Nominal Share Price (log)	0.022***	0.006***	-0.003***	-0.009***	0.032***	0.015***	-0.001	-0.011***
	(27.56)	(6.11)	(-3.05)	(-9.69)	(38.98)	(14.55)	(-0.67)	(-10.38)
Daily Return (log)	-0.003***	-0.002***	-0.001***	0.000	-0.004***	-0.003***	-0.002***	-0.001***
	(-28.20)	(-15.93)	(-6.95)	(0.67)	(-31.94)	(-15.64)	(-14.11)	(-3.30)
Traded Volume (log)	0.069***	0.031***	0.000	-0.024***	0.072***	0.033***	0.002***	-0.026***
	(201.99)	(78.49)	(0.03)	(-54.33)	(196.72)	(75.05)	(3.82)	(-57.00)
Total Market Cap (log)	-0.004***	-0.008***	-0.009***	-0.007***	-0.003***	-0.007***	-0.010***	-0.007***
	(-4.25)	(-8.22)	(-8.81)	(-6.87)	(-2.91)	(-6.88)	(-9.35)	(-6.53)
Proportional Spread	0.015***	0.027***	0.033***	0.035***	0.018***	0.031***	0.035***	0.037***
	(45.35)	(71.30)	(80.55)	(81.04)	(49.25)	(71.46)	(84.90)	(82.46)
Volatility	-0.053***	-0.018***	0.024***	0.060***	-0.056***	-0.021***	0.023***	0.065***
	(-28.49)	(-9.08)	(11.94)	(27.15)	(-29.95)	(-9.73)	(11.41)	(27.60)
Beta with CSI300	-0.004***	-0.004***	-0.003***	-0.003***	-0.004***	-0.004***	-0.004***	-0.003***
	(-17.40)	(-15.78)	(-12.40)	(-9.28)	(-17.20)	(-14.29)	(-13.97)	(-9.56)
R2 with CSI300	0.009***	-0.002***	-0.013***	-0.020***	0.009***	-0.002***	-0.013***	-0.021***
	(24.68)	(-5.94)	(-32.47)	(-47.45)	(25.05)	(-4.71)	(-32.84)	(-47.19)
Idiosyncratic Volatility	0.026***	-0.015***	-0.053***	-0.080***	0.027***	-0.016***	-0.055***	-0.087***
	(14.09)	(-7.87)	(-26.68)	(-37.03)	(14.53)	(-7.61)	(-27.78)	(-37.73)
Jump	-0.002***	-0.001***	-0.001***	-0.000**	-0.002***	-0.001***	-0.000**	0.000
	(-14.92)	(-8.54)	(-4.33)	(-2.37)	(-13.62)	(-8.47)	(-2.06)	(0.44)
Observations	165,083	165,083	165,083	165,083	163,567	163,567	163,567	163,567
R-squared	0.266	0.203	0.306	0.360	0.264	0.207	0.354	0.373
Symbol fixed effect	YES	YES	YES	YES	YES	YES	YES	YES
Date fixed effect	YES	YES	YES	YES	YES	YES	YES	YES

Robust t-statistics in parentheses
*** p<0.01, ** p<0.05, * p<0.1

and quickly corrected by arbitrageurs.

Second, short-selling constraints impede efficiency. The *Margin Trading Target* dummy is strongly negative across all horizons (-0.005 to -0.009, $p < 0.01$). This confirms that when short-selling is allowed, sophisticated traders eliminate mispricing more effectively (Diamond and Verrecchia, 1987).

Third, institutional capital improves price discovery. Both *Net Institutional Flow* and *Northbound Holdings* exhibit consistently negative coefficients, indicating that informed trading reduces predictable noise. Finally, leveraged speculation, proxied by *Margin Buy Balance*, positively predicts returns (0.004 to 0.011***), implying that leverage-induced sentiment introduces persistent behavioral biases.

4.6 Causal Identification: A Quasi-Natural Experiment

While the cross-sectional regressions in Section 4.5 suggest that limits to arbitrage play a role, unobserved heterogeneity could still drive the results. To establish a clean causal link between short-selling constraints and high-frequency predictability, the staggered inclusion of stocks into the Margin Trading and Short Selling (MTSS) list is exploited as a quasi-natural experiment.

Table 5: Investor Composition and Market Frictions

VARIABLES	(1) Ridge 5s	(2) Ridge 15s	(3) Ridge 30s	(4) Ridge 60s	(5) RF 5s	(6) RF 15s	(7) RF 30s	(8) RF 60s
Retail Trade Ratio	0.016***	0.006***	-0.002***	-0.007***	0.017***	0.007***	-0.001***	-0.007***
	(86.06)	(26.88)	(-8.67)	(-29.60)	(82.91)	(28.01)	(-6.14)	(-28.12)
Net Institutional Flow	-0.001***	-0.001***	-0.002***	-0.002***	-0.001***	-0.002***	-0.002***	-0.002***
	(-8.00)	(-10.96)	(-11.40)	(-11.25)	(-8.66)	(-11.00)	(-12.72)	(-10.48)
Margin Trading Target	-0.005***	-0.008***	-0.009***	-0.008***	-0.005***	-0.010***	-0.010***	-0.009***
	(-17.35)	(-24.85)	(-25.34)	(-22.47)	(-17.41)	(-26.82)	(-28.74)	(-23.93)
Northbound Holdings	-0.006***	-0.009***	-0.007***	-0.004***	-0.006***	-0.009***	-0.007***	-0.005***
	(-16.00)	(-22.88)	(-19.87)	(-12.97)	(-15.57)	(-22.93)	(-20.34)	(-12.45)
Margin Buy Balance	0.005***	0.010***	0.010***	0.008***	0.004***	0.011***	0.010***	0.008***
	(12.95)	(22.30)	(22.61)	(16.82)	(10.93)	(23.16)	(23.70)	(15.99)
Observations	165,083	165,083	165,083	165,083	163,567	163,567	163,567	163,567
R-squared	0.301	0.214	0.313	0.368	0.298	0.218	0.361	0.381
Symbol fixed effect	YES	YES	YES	YES	YES	YES	YES	YES
Date fixed effect	YES	YES	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES	YES	YES

Robust t-statistics in parentheses
*** p<0.01, ** p<0.05, * p<0.1

4.6.1 Experimental Design and Data Construction

In the Chinese market, stocks are added to the MTSS list only when they meet specific liquidity and capitalization thresholds determined by the exchanges. This inclusion creates an exogenous shock to the cost of arbitrage: prior to inclusion, short selling is strictly prohibited; post inclusion, qualified investors can borrow shares to short.

To address potential selection bias, a matched sample is constructed based on the exact eligibility criteria enforced by the exchanges. For each stock added to the MTSS list (Treatment Group), a control stock (Control Group) is identified that is not on the list but shares the most similar "eligibility score." This score is calculated based on the exchange's official formula, weighting the average market capitalization and trading volume over the preceding 60 trading days. A 1:1 nearest neighbor matching approach without replacement is employed to ensure the treatment and control groups share parallel trends in liquidity and volatility prior to the event.

The analysis focuses on the event window of $[-12, +12]$ weeks centered on the inclusion date and estimates the following difference-in-differences (DID) specification:

$$R^2_{i,t} = \alpha + \beta(\text{Treat}_i \times \text{Post}_{i,t}) + \gamma X_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t} \tag{8}$$

where Treat_i is a dummy variable for the treatment group, and $\text{Post}_{i,t}$ equals one after the inclusion date. The coefficient β captures the causal impact of relaxing short-selling constraints on predictability.

4.6.2 Baseline Results

Table 6 presents the estimation results for both linear (Ridge) and non-linear (Random Forest) prediction models across all horizons.

Table 6: Impact of Short-Selling Deregulation on Predictability (DID Analysis)

VARIABLES	(1) Ridge 5s	(2) Ridge 15s	(3) Ridge 30s	(4) Ridge 60s	(5) RF 5s	(6) RF 15s	(7) RF 30s	(8) RF 60s
Treat × Post	-0.007***	-0.009***	-0.008***	-0.006***	-0.009***	-0.012***	-0.009***	-0.008***
	(-7.56)	(-9.17)	(-7.62)	(-5.49)	(-8.71)	(-10.21)	(-8.52)	(-6.24)
Observations	20,699	20,699	20,699	20,699	20,526	20,526	20,526	20,526
R-squared	0.259	0.161	0.240	0.291	0.242	0.160	0.298	0.297
Symbol fixed effect	YES	YES	YES	YES	YES	YES	YES	YES
Date fixed effect	YES	YES	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES	YES	YES

Robust t-statistics in parentheses
*** p<0.01, ** p<0.05, * p<0.1

The results provide compelling evidence that relaxing arbitrage constraints reduces predictability. The coefficients on the interaction term (Treat × Post) are negative and statistically significant at the 1% level across all specifications.

Regarding economic magnitude, for the Random Forest model, the inclusion leads to a reduction in out-of-sample R^2 ranging from 0.8% to 1.2% (absolute percentage points). Given the average baseline predictability, this represents a relative decline of approximately 5% to 10%. In terms of robustness, the consistency between Ridge (Columns 1-4) and Random Forest (Columns 5-8) confirms that this effect is structural and not an artifact of specific machine learning architectures.

This finding supports the hypothesis that short sellers act as sophisticated arbitrageurs. Once allowed to enter the market, they rapidly incorporate negative information into prices, thereby eliminating the serial correlations and inefficiencies that the models otherwise exploit.

4.6.3 Dynamic Effects and Parallel Trends

A key identifying assumption of the DID framework is that the treatment and control groups would have followed parallel trends in the absence of the shock. To validate this, a dynamic DID model is estimated:

$$R^2_{i,t} = \alpha + \sum_{k=-12}^{12} \beta_k D^k_{i,t} + \gamma X_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t} \tag{9}$$

Figure 9 plots the estimated coefficients β_k and their 95% confidence intervals.

The figure reveals two crucial patterns. First, regarding pre-event validity, for $k < 0$, the coefficients are statistically indistinguishable from zero and exhibit no systematic

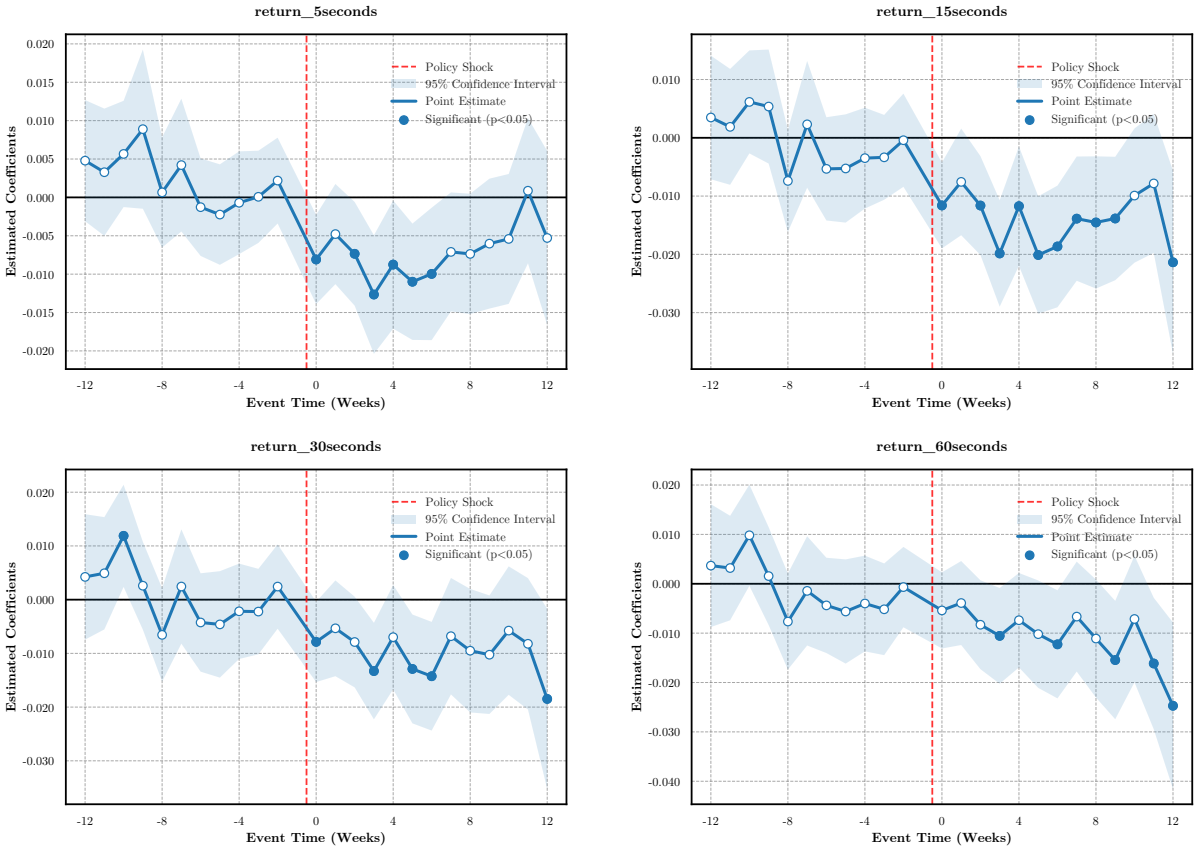


Figure 9: Dynamic Impact of Short-Selling Deregulation on Predictability

Note: The figure plots the dynamic coefficients β_k from the event study specification. The dashed vertical line at $t = 0$ represents the inclusion date. The blue solid line represents the point estimates, and the shaded area indicates the 95% confidence interval. The "cliff-like" drop at $t = 0$ confirms the immediate causal effect of short-selling availability on reducing return predictability.

trend. This confirms that the matching procedure successfully eliminated pre-existing differences between the treatment and control groups, validating the parallel trends assumption. Second, concerning post-event impact, immediately following the inclusion ($k \geq 0$), a sharp and persistent decline in the coefficients is observed. This "cliff-like" drop indicates that the improvement in market efficiency (reduction in predictability) occurs almost instantaneously as short-selling channels open. The effect remains stable over the subsequent 12 weeks, suggesting that the entry of short sellers leads to a permanent shift in the information environment rather than a temporary reaction.

In summary, the results from this quasi-natural experiment provide robust causal evidence that the high predictability observed in the Chinese market is, at least in part, a consequence of regulatory limits to arbitrage.

5 Trading Strategy Performance

Building on the predictability documented in the previous section, we now evaluate the economic significance of our high-frequency signals. We implement an execution optimization strategy that systematically exploits short-term return predictability to improve trade execution prices. This approach aligns with the practical constraints of institutional investors in the Chinese market, where transaction costs and market impact are primary concerns.

5.1 Strategy Design and Execution Framework

Our trading strategy translates predictive model outputs into actionable execution decisions. Rather than constructing a high-turnover long-short portfolio—which would likely succumb to transaction costs given the T+1 trading rule—we focus on execution optimization (e.g., algorithmic trading). The objective is to optimize the timing of trades for a large institutional order to minimize implementation shortfall relative to standard benchmarks.

5.1.1 Execution Optimization Logic

The strategy operates under the premise that an investor needs to execute a predetermined volume V over a given time horizon T. We employ an adaptive delay mechanism driven by our machine learning signals. When the model predicts a favorable price movement (e.g., a positive return for a buy order), the algorithm delays execution to capture the lower price. Conversely, if an adverse movement is predicted, the algorithm executes immediately to avoid price slippage.

Formally, for each prediction horizon $h \in \{5s, 15s, 30s, 60s\}$, we generate a composite signal $\hat{r}_{t+h}^{ensemble}$ by aggregating forecasts from our machine learning models. We then apply a dynamic thresholding rule:

$$\text{Action}_t = \begin{cases} \text{Aggressive Execution} & \text{if } \hat{r}_{t+h}^{ensemble} < \theta_{buy} \text{ (Buy Order)} \\ \text{Passive/Delayed Execution} & \text{if } \hat{r}_{t+h}^{ensemble} > \theta_{buy} \text{ (Buy Order)} \end{cases} \tag{10}$$

where θ_{buy} is a threshold optimized over a rolling training window to maximize historical execution performance. A similar logic applies to sell orders. To manage inventory risk, we impose a maximum delay constraint, ensuring that the full order is completed within the specified horizon regardless of the signal.

5.1.2 Parameterization and Calibration

The strategy involves several key hyperparameters calibrated to balance signal capture against execution risk. The maximum delay window is set to 10 intervals, allowing sufficient time for short-term inefficiencies to resolve. Thresholds θ are updated daily using a 60-day rolling window, ensuring the strategy adapts to changing market volatility and liquidity conditions. We synthesize signals across horizons using a linear ensemble, weighting predictions by their historical information ratios. This multi-horizon approach robustly captures both ultra-high-frequency microstructure noise (5s) and slightly longer-term order flow momentum (60s).

5.2 Execution Performance Analysis

We evaluate strategy performance using execution quality metrics standard in the algorithmic trading literature. Our primary benchmarks are the Time-Weighted Average Price (TWAP) and Volume-Weighted Average Price (VWAP).

5.2.1 Execution Quality

Table 7 presents the execution performance across different prediction horizons. The results demonstrate statistically and economically significant improvements. The 60-second horizon strategy achieves the strongest performance, yielding an average improvement of 5.1 basis points (bps) over TWAP and 4.3 bps over VWAP. Even at the 5-second horizon, the strategy delivers a statistically significant 2.3 bps improvement over TWAP.

The magnitude of these improvements is substantial in the context of institutional execution. For a large fund turnover of RMB 10 billion annually, a 5 bps improvement translates to RMB 5 million in direct cost savings. The information ratios, calculated as the mean improvement divided by its standard deviation, consistently exceed 0.8 and reach 1.51 for the 60-second strategy, indicating that the excess returns are stable and not driven by outliers.

The ensemble strategy, which combines signals across all horizons, achieves a robust 4.8 bps improvement over TWAP with an information ratio of 1.42. This suggests that diversifying across signal horizons effectively mitigates model-specific noise, providing a more reliable execution algorithm.

5.2.2 Robustness Across Market Conditions

We further examine the consistency of strategy performance across varying market conditions. First, sorting stocks by market capitalization reveals that execution improvements are pervasive but slightly stronger in mid-cap stocks. This aligns with our earlier finding that mid-cap stocks exhibit a favorable balance of liquidity (allowing execution) and

Table 7: Execution Strategy Performance Analysis

Horizon	TWAP Improvement (bps)	VWAP Improvement (bps)	Signal Frequency (%)	Infor
5-second	2.3***	1.8***	12.4	
15-second	3.1***	2.6***	15.8	
30-second	4.2***	3.5***	18.9	
60-second	5.1***	4.3***	22.1	
Ensemble Strategy	4.8***	4.0***	20.3	

*Note: This table reports execution performance improvements relative to passive TWAP and VWAP benchmarks. Performance is measured in basis points (bps). Signal frequency represents the percentage of trading intervals where the strategy generates active signals. Information ratios are calculated as mean improvement divided by standard deviation. *** indicates significance at the 1% level based on bootstrap confidence intervals.*

inefficiency (allowing prediction).

Second, we analyze performance across volatility regimes. The strategy delivers significant improvements in both high and low volatility periods, although the absolute magnitude of savings is larger during volatile markets. This counter-cyclical property is particularly valuable, as execution costs typically spike during market turbulence.

Third, we document distinct intraday patterns. Execution improvements are most pronounced during the opening (9:30–10:00) and closing (14:30–15:00) sessions. These periods correspond to peak retail activity and elevated volume, confirming that our models effectively exploit the predictable order flow imbalances generated by noise traders.

5.3 Transaction Costs and Implementation

A realistic assessment of profitability must account for transaction costs and market impact. We model costs using a calibrated square-root impact model (**?**), adjusted for the Chinese market structure.

5.3.1 Cost-Adjusted Performance

Table 8 reports the net performance under three transaction cost scenarios: conservative (15 bps), realistic (10 bps), and optimistic (5 bps). Under the realistic scenario, which accounts for bid-ask spreads, commissions, and market impact, the ensemble strategy retains a net TWAP improvement of 3.6 bps. This result confirms that the statistical predictability documented in Section 4 survives rigorous transaction cost scrutiny.

The break-even analysis indicates that the strategy remains profitable as long as the executed volume does not exceed approximately 12% of the stock's daily volume. Beyond this point, the market impact of the trade begins to erode the signal advantage. This

capacity constraint is typical for high-frequency strategies but allows for deployment on significant institutional capital bases.

Table 8: Performance Under Different Transaction Cost Scenarios

Cost Scenario (basis points)	**Net TWAP Improvement (bps)**	**Net VWAP Improvement (bps)**	**Break-even Volume (%)**
Conservative (15 bps)	2.8***	2.3***	8.5
Realistic (10 bps)	3.6***	3.1***	12.8
Optimistic (5 bps)	4.4***	3.9***	18.6
Ensemble Strategy	3.5***	3.0***	12.2

*Note: This table shows net execution improvements after transaction costs. Break-even volume represents the maximum percentage of daily volume that can be executed before strategy profitability is eliminated. *** indicates significance at the 1% level.*

5.4 Economic Implications

Our findings have broader implications for market efficiency and regulation. The persistence of economically significant predictability suggests that the Chinese market, despite its rapid modernization, retains structural inefficiencies. These are likely driven by the dominance of retail investors and the T+1 trading rule, which limits the ability of arbitrageurs to correct mispricing immediately.

For practitioners, the results highlight the necessity of intelligent execution. In a market where alpha is decaying, reducing execution costs by 3–5 bps is equivalent to generating significant additional alpha. For regulators, the findings suggest that while high-frequency trading is often viewed with skepticism, algorithmic execution tools can improve market quality by reducing price dispersion and facilitating efficient risk transfer for institutional investors.

6 Robustness Checks

7 Conclusion

A Additional Tables and Figures

This appendix provides additional empirical results, robustness checks, and supplementary analysis that support the main findings but are too detailed for the main text. Extended model performance statistics, sector-specific analysis, and detailed feature definitions are presented.

A.1 The Chinese Stock Market Structure

The Chinese stock market provides a unique laboratory for testing high-frequency predictability due to several distinctive institutional features that differentiate it from the U.S. market analyzed in Aït-Sahalia et al. (2022).

Market Organization. The analysis covers stocks traded on both the Shanghai Stock Exchange (SSE) and Shenzhen Stock Exchange (SZSE). These exchanges operate fully electronic trading systems with continuous trading from 9:30 AM to 3:00 PM, with a lunch break from 11:30 AM to 1:00 PM. The market has grown to become the world's second-largest equity market by capitalization.

Price Limits and Regulatory Constraints. A critical feature of the Chinese market is the strict daily price limit system. Stocks are subject to a ±10% daily price limit (±5% for ST/*ST stocks), creating hard boundaries on price movements. This mechanism can fundamentally alter order flow dynamics, creating non-linearities and potential predictability patterns not observable in U.S. markets, especially as prices approach these limits. The "magnet effect" near price limits may create state-dependent predictability that is explored in the analysis.

Investor Composition. Unlike the institutionally-dominated U.S. market, the Chinese market is characterized by a high proportion of retail investors, who account for approximately 80% of trading volume. This retail dominance may lead to different patterns of noise trading, herding behavior, and information processing, which could be exploited by or affect high-frequency trading strategies.

Trading Mechanisms. The T+1 settlement rule and historical restrictions on short-selling shape intraday inventory management concerns of market participants, which could indirectly influence short-term predictability patterns.

A.2 Level-2 Data Structure and Processing

This section details the structure of the high-frequency dataset and the preprocessing methodology employed. The dataset comprises two primary components: transaction

(tick-by-tick) data and limit order book (LOB) snapshots.

A.2.1 Limit Order Book Data

Table 9 illustrates the structure of the Level-2 order book snapshots, using Hunan Development (SZ.000722) as a representative example. The data captures the state of the limit order book at discrete intervals, providing depth information for the top 10 bid and ask levels.

Table 9: Level-2 Order Book Data for Hunan Development (SZ.000722)

Panel A: Data Sample

	Bid Side						Ask Side					
TRIGGER_TIME	B1	BQ1	B2	BQ2	B3	BQ3	A1	AQ1	A2	AQ2	A3	AQ3
20210723102021464	6.81	49300	6.80	109000	6.79	52500	6.82	36100	6.83	51300	6.84	11900
20210723102024456	6.81	49300	6.80	109000	6.79	52500	6.82	36100	6.83	51300	6.84	11900
20210723102030456	6.81	49100	6.80	109000	6.79	52500	6.82	36100	6.83	51300	6.84	11900
20210723102033464	6.81	49100	6.80	109000	6.79	52500	6.82	36100	6.83	51300	6.84	11900
20210723102036456	6.81	49100	6.80	108700	6.79	52500	6.82	36100	6.83	51300	6.84	11900
20210723102039456	6.81	42800	6.80	107700	6.79	52500	6.82	36100	6.83	51300	6.84	11900
20210723102042456	6.81	44000	6.80	112700	6.79	52500	6.82	35900	6.83	51300	6.84	11900
20210723102045456	6.81	42200	6.80	112700	6.79	55500	6.82	36900	6.83	51300	6.84	11900
20210723102048456	6.81	42800	6.80	112700	6.79	55500	6.82	36900	6.83	51300	6.84	11900
20210723102051464	6.81	47800	6.80	112700	6.79	55500	6.82	36900	6.83	51300	6.84	11900

Panel B: Field Descriptions

Field	Unit/Format	Description
TRIGGER_TIME	YYYYMMDDHHMMSSfff	Timestamp when the market quote snapshot was triggered (nanosecond resolution)
B1–B10	RMB (tick=0.01)	Top 10 bid price levels (B1 = best bid price)
BQ1–BQ10	Shares (×100)	Aggregate order quantity at each bid price level
A1–A10	RMB (tick=0.01)	Top 10 ask price levels (A1 = best ask price)
AQ1–AQ10	Shares (×100)	Aggregate order quantity at each ask price level

Note: This table presents Level-2 order book snapshots for Hunan Development (SZ.000722) on July 23, 2021. Panel A shows a sample of the data with the top 3 price levels displayed. The full dataset includes the top 10 price levels on each side (B1–B10, BQ1–BQ10, A1–A10, AQ1–AQ10). Panel B describes all fields in the dataset. TRIGGER_TIME represents the timestamp when the snapshot was triggered by the exchange system. Prices are in RMB with a minimum tick size of 0.01, and quantities are in shares (multiples of 100 per exchange rules).

Variable Definitions. The variables are defined as follows: *TRIGGER_TIME* denotes the exchange timestamp with nanosecond precision (format: YYYYMMDDHHMMSSfff). *B1–B10 (A1–A10)* represent the prices for the top 10 bid (ask) levels, ordered by price priority, where prices are in RMB with a tick size of 0.01. *BQ1–BQ10 (AQ1–AQ10)* indicate the aggregate volume (in shares) at the corresponding bid (ask) price levels.

A.2.2 Transaction Data

Table 10 presents sample transaction records. This dataset captures all market events, including order submissions, cancellations, and trade executions.

Table 10: Transaction Data for Hunan Development (SZ.000722)

Panel A: Data Sample

SEQID	TYPE	TRIGGER_TIME	PRICE	VOLUME	SIDE	ORDER_TYPE	ORDER_NO	BID_NO	ASK_NO
66313678	E	20210723102019983	6.91	1000	2.0	2	13272300	0	0
66330153	E	20210723102021108	6.78	1200	1.0	2	13275761	0	0
66454298	C	20210723102029245	0.00	2600				68142	0
66455174	E	20210723102029355	6.81	1200	2.0	2	13300244	0	0
66455175	T	20210723102029355	6.81	100				8650378	13300244
66455176	T	20210723102029355	6.81	500				8662273	13300244
66455177	T	20210723102029356	6.81	600				8672886	13300244
66461449	E	20210723102029867	6.81	1000	1.0	2	13301598	0	0
66471078	C	20210723102030578	0.00	11100				0	12136253
66507429	E	20210723102033114	6.67	1000	1.0	2	13310449	0	0
66514454	C	20210723102033585	0.00	300				3555450	0

Panel B: Field Descriptions

Field	Unit/Format	Description
SEQID	Integer	Unique sequential identifier for each event
TYPE	Character	Type of market event: E (order entry), C (order cancellation), T (trade execution)
TRIGGER_TIME	YYYYMMDDHHMMSSfff	Timestamp when the event was triggered (nanosecond resolution)
PRICE	RMB (tick=0.01)	Price level of the order or trade
VOLUME	Shares (×100)	Number of shares involved in the order or trade
SIDE	Integer	Order/Trade side: 1 (buy order or buyer-initiated trade), -1 (sell order or seller-initiated trade)
ORDER_TYPE	Integer	Type of order: 1 (market order), 2 (limit order), etc.
ORDER_NO	Integer	Unique order identifier assigned by the exchange
BID_NO	Integer	Order number on bid side (used for tracking order cancellations)
ASK_NO	Integer	Order number on ask side (used for tracking order cancellations)

Note: This table presents transaction records for Hunan Development (SZ.000722) on July 23, 2021. Panel A shows a complete sample of the data with all available fields. Panel B describes all fields in the dataset. TRIGGER_TIME represents the timestamp when the event was triggered by the exchange system. For TYPE="E" (entry), SIDE indicates order direction; for TYPE="T" (trade), SIDE is determined by the Lee and Ready (1991) algorithm. For TYPE="C" (cancellation), BID_NO or ASK_NO identifies which order is being cancelled. Prices are in RMB with a minimum tick size of 0.01, and volumes are in shares (multiples of 100 per exchange rules).

Variable Definitions. Key variables include: *SEQID*, a unique sequential identifier for market events; *TYPE*, an event type indicator where 'E' denotes Order Entry, 'C' Cancellation, and 'T' Trade Execution; and *TRIGGER_TIME*, the exchange timestamp with nanosecond precision. *PRICE* and *VOLUME* denote the transaction or order price (RMB) and quantity of shares, respectively. *SIDE* indicates trade initiation direction, taking a value of 1 for buyer-initiated and -1 for seller-initiated trades. *ORDER_TYPE* classifies the order (e.g., 1 for market orders, 2 for limit orders). *ORDER_NO* is the unique order identifier, while *BID_NO* and *ASK_NO* identify the specific bid and ask orders involved in a trade.

A.2.3 Data Preprocessing Methodology

A rigorous preprocessing framework is implemented to ensure data integrity, temporal alignment, and the elimination of look-ahead bias.

Transaction Data Processing. The processing of transaction data focuses on reconstructing trade execution details and classifying trade direction. The procedure involves several steps. First, for *Order Reconstruction*, order entry records (TYPE='E') are isolated to construct separate bid and ask order repositories, indexed by `ORDER_NO`. Zero-price entries are treated as missing and excluded.

Second, *Trade Matching* is performed by matching execution records (TYPE='T') with their corresponding originating orders using `BID_NO` and `ASK_NO`. This allows for the recovery of precise entry prices and volumes for both sides of each transaction.

Third, *Trade Classification* is determined based on time priority. In contrast to the Lee and Ready (1991) algorithm, which infers trade direction from price and quote data, the Chinese Level-2 data allows for exact classification. A trade is classified as buyer-initiated (SIDE=1) if the ask order timestamp precedes the bid order timestamp, and seller-initiated (SIDE=-1) otherwise.

Fourth, to ensure *Temporal Alignment and Look-Ahead Bias Mitigation*, a forward-shift mechanism is applied to the timestamps. Raw nanosecond timestamps are converted to second-level frequency by rounding up to the next integer second:

$$\text{TIME}_t = \left\lfloor \frac{\text{TRIGGER_TIME}_t \bmod 10^9}{10^3} \right\rfloor + 1$$

This transformation ensures that information generated within second t is only available for predictions at $t + 1$. Finally, missing price data in matched orders is *imputed* using the executed trade price as a conservative proxy.

Order Book Snapshot Processing. Snapshot data undergoes a similar standardization process. *Timestamp Standardization* involves forward-shifting timestamps by one second to align with the transaction data and maintain causality. *Data Aggregation* structures price and volume data for the top 10 levels into vector format for computational efficiency. In instances of simultaneous timestamps, *De-duplication* retains the most recent snapshot to reflect the latest market state.

Synchronization and Missing Data Handling. Given the asynchronous nature of high-frequency data, all observations are synchronized to a uniform time grid. A reference timeline is constructed containing all integer seconds during continuous trading hours (09:30:00–11:30:00 and 13:00:00–14:57:00), resulting in 14,400 observations per day. Regarding *Price Variables*, missing values are forward-filled using the most recent observation, consistent with the martingale property of prices at high frequencies. For *Flow Variables*, variables representing activity (e.g., volume, trade counts) are zero-filled for intervals with no activity. This framework yields a synchronized, second-level dataset that preserves the microstructure characteristics of the market while ensuring suitability for predictive modeling.

A.3 Factor Definitions

The following formulas provide supplementary details for the complex predictors.

Table 11: Detailed Factor Definitions

Type	Alias	Definition	Formula
Volume and Trading Intensity	Breadth	Number of transactions in the lookback interval	$\left\lvert \mathcal{D}^{\text{txn}} \cap \text{Int}^{\text{back}}(T, \Delta_1, \Delta_2) \right\rvert$
	Immediacy	Average time between successive transactions	$\frac{\Delta_2 - \Delta_1}{\text{Breadth}+1}$
	VolumeAll	Total number of shares transacted	$\sum_{t \in \mathbf{I}} V_t$
	VolumeAvg	Average transaction size	$\frac{\text{VolumeAll}}{\text{Breadth}+1}$
	VolumeMax	Maximum single transaction volume	$\max\{V_t : t \in \mathbf{I}\}$
	VolumeStd	Standard deviation of transaction volumes	$\sqrt{\frac{1}{n-1} \sum_{t \in \mathbf{I}} (V_t - \text{VolumeAvg})^2}$
	VolumeSkew	Asymmetry in transaction volume distribution	$\frac{\frac{1}{n} \sum_{t \in \mathbf{I}} (V_t - \text{VolumeAvg})^3}{[\text{VolumeStd}]^3}$
	VolumeSideSum	Signed volume flow	$\sum_{t \in \mathbf{I}} (V_t \cdot \text{Dir}_t)$
	Price_T	Most recent transaction price	$P_{\max\{t \in \mathcal{D}^{\text{txn}} : t < T\}}$
Return and Price Dynamics	Lambda	Price change relative to total volume	$\frac{P_{\max(\mathbf{I})} - P_{\min(\mathbf{I})}}{\text{VolumeAll}+100}$
	PastReturn	Historical return over the lookback window	$1 - \frac{\text{Average}[P_t : t \in \mathbf{I}]}{P_{\max(\mathbf{I})}}$
	PriceHL	High-low price range in log form	$\log\left(\frac{P_{\max(\mathbf{I})}+0.01}{P_{\min(\mathbf{I})}+0.01}\right)$
	PriceReturn	Simple return over the window	$\log\left(\frac{P_{\max(\mathbf{I})}+0.01}{P_{\min(\mathbf{I})}+0.01}\right)$
Order Imbalance and Trading Direction	TxnImbalance	Signed volume imbalance	$\frac{\sum_{t \in \mathbf{I}} (V_t \cdot \text{Dir}_t)}{\text{VolumeAll}+100}$
	LobImbalance	Limit order book imbalance at best bid/ask	$\text{Avg}\left[\frac{S_t^a - S_t^b}{S_t^a + S_t^b + 100}\right]$
	BuyRatio	Proportion of buy-initiated volume	$\frac{\sum_{t \in \mathbf{I}} (1+\text{Dir}_t) V_t}{\sum_{t \in \mathbf{I}} P_t V_t}$
	SellRatio	Proportion of sell-initiated volume	$-\frac{\sum_{t \in \mathbf{I}} (1-\text{Dir}_t) V_t}{\sum_{t \in \mathbf{I}} P_t V_t}$
Trading Cost and Spread	QuotedSpread	Average proportional bid-ask spread	$\text{Avg}\left[\frac{P_t^a - P_t^b}{(P_t^a + P_t^b)/2}\right]$

Table 11: Detailed Factor Definitions (Continued)

Type	Alias	Definition	Formula
	EffectiveSpread	Dollar-weighted effective spread	$\frac{\sum_{t\in\mathbf{I}}[\log(P_t/P_T)\cdot \mathrm{Dir}_t\cdot P_tV_t]}{\sum_{t\in\mathbf{I}}(P_tV_t)+100}$
	AutoCov	Autocovariance of transaction returns	$\frac{1}{\mathrm{Breadth}+1}\sum \Delta\log P_t \Delta\log P_{t-1}$
Order Book Depth Features	BuyIntention	Intensity of aggressive buying	$\frac{\sum_{t\in\mathbf{I}} P_t^b V_t}{\sum_{t\in\mathbf{I}} P_tV_t} - 1$
	SellIntention	Intensity of aggressive selling	$1 - \frac{\sum_{t\in\mathbf{I}} P_t^a V_t}{\sum_{t\in\mathbf{I}} P_tV_t}$
	AskBidAmountRatio	Imbalance in aggregate depth	$\log\left(\frac{\sum P_T^{a,i}S_T^{a,i}+100}{\sum P_T^{b,i}S_T^{b,i}+100}\right)$
	logQuoteSlope	Price-depth relationship	See Eq. 11
	VWAPSlope	VWAP comparison	See Eq. 12
	SnapAmountSum	Total liquidity across depth	$\sum(P_T^{a,i}S_T^{a,i} + P_T^{b,i}S_T^{b,i})$
	SnapAmountStd	Dispersion of liquidity distribution	See Eq. 13
	SnapAmountSkew	Asymmetry in liquidity distribution	See Eq. 14
	SnapMaxSlope	Steepest price gradient	See Eq. 15
	WapBalance	Weighted average price imbalance	$\frac{\mathrm{VWAP}^b-\mathrm{VWAP}^a}{(\mathrm{VWAP}^b+\mathrm{VWAP}^a)/2}$
	BuyIntentionMax	Maximum aggressive buying intensity	$\max_{t\in\mathbf{I}}\left(\frac{P_t^bV_t}{P_tV_t} - 1\right)$
	SellIntentionMax	Maximum aggressive selling intensity	$\max_{t\in\mathbf{I}}\left(1 - \frac{P_t^aV_t}{P_tV_t}\right)$
	PriceBias	Deviation from midpoint	$\frac{P_T-\mathrm{Mid}_T}{\mathrm{Mid}_T}$

Supplementary Formulas.

$$\mathrm{logQuoteSlope}(T) = \sum_{i=1}^{10} \frac{\log(P_T^{a,i} + 10^{-6}) - \log(P_T^{b,i} + 10^{-6})}{\log(S_T^{b,1} + 1) - \log(S_T^{a,1} + 1) + 10^{-6}}. \tag{11}$$

$$\mathrm{VWAPSlope}(T) = \frac{\mathrm{VWAP}^a(T) - P_T^{a,1} + 10^{-6}}{P_T^{b,1} - \mathrm{VWAP}^b(T) + 10^{-6}}, \tag{12}$$

where $\mathrm{VWAP}^a(T) = \sum_{i=1}^{10} P_T^{a,i} S_T^{a,i} / \sum_{i=1}^{10} S_T^{a,i}$ is the volume-weighted average ask price across all 10 ask levels.

$$\mathrm{SnapAmountStd}(T) = \sqrt{\frac{1}{19} \sum_{i=1}^{10} \left[(P_T^{a,i} S_T^{a,i} - \bar{A})^2 + (P_T^{b,i} S_T^{b,i} - \bar{A})^2 \right]}, \tag{13}$$

where $\bar{A} = \mathrm{SnapAmountSum}(T)/20$ is the mean amount across all levels.

$$\mathrm{SnapAmountSkew}(T) = \frac{\frac{1}{20} \sum_{i=1}^{10} \left[(P_T^{a,i} S_T^{a,i} - \bar{A})^3 + (P_T^{b,i} S_T^{b,i} - \bar{A})^3 \right]}{[\mathrm{SnapAmountStd}(T)]^3}. \tag{14}$$

$$\mathrm{SnapMaxSlope}(T) = \max_{i=1,\ldots,9} \left\{ \frac{P_T^{a,i+1} - P_T^{a,i}}{P_T^{a,i}}, \frac{P_T^{b,i} - P_T^{b,i+1}}{P_T^{b,i}} \right\}. \tag{15}$$

A.4 Feature Engineering and Statistical Transformations

To capture the dynamic evolution and distributional properties of market activity, a temporal expansion strategy is employed that enriches cross-sectional predictors with statistical features computed over rolling windows. Formally, for any base predictor X and lookback span (Δ_1, Δ_2) with $\Delta_1 = 0$, let $\{X_t : t \in \mathrm{Int}^{\mathrm{back}}(T, 0, \Delta_2)\}$ denote the sequence of observations in the lookback window. The following three moments are computed:

$$\mathrm{Mean}[X](T, \Delta_2) = \frac{1}{n} \sum_{t \in \mathrm{Int}^{\mathrm{back}}(T,0,\Delta_2)} X_t, \tag{16}$$

$$\mathrm{Std}[X](T, \Delta_2) = \sqrt{\frac{1}{n-1} \sum_{t \in \mathrm{Int}^{\mathrm{back}}(T,0,\Delta_2)} \Big(X_t - \mathrm{Mean}[X](T, \Delta_2) \Big)^2}, \tag{17}$$

$$\mathrm{Skew}[X](T, \Delta_2) = \frac{\frac{1}{n} \sum_{t \in \mathrm{Int}^{\mathrm{back}}(T,0,\Delta_2)} \Big(X_t - \mathrm{Mean}[X](T, \Delta_2) \Big)^3}{\Big(\mathrm{Std}[X](T, \Delta_2) \Big)^3}, \tag{18}$$

where $n = |\mathrm{Int}^{\mathrm{back}}(T, 0, \Delta_2)|$ is the number of observations in the window.

These transformations are applied to 16 base predictors selected for their temporal variability. The first group of 8 transaction-based variables derived from tick-by-tick trade data includes: VolumeSideSum, Price_T, BuyRatio, SellRatio, BuyIntention, SellIntention, VolumeStd, and VolumeSkew. The second group of 8 order book snapshot variables includes: AskBidAmountRatio, SnapMaxSlope, VWAPSlope, SnapAmountSum, SnapAmountStd, SnapAmountSkew, WapBalance, and logQuoteSlope. For these 16 predictors, computing mean, standard deviation, and skewness over 8 lookback windows yields $16 \times 3 \times 8 = 384$ statistical features. Combined with the 17 direct predictors (each evaluated over 8 windows, yielding $17 \times 8 = 136$ features), the total feature space comprises $(16 \times 3 + 17) \times 8 = 520$ predictors at each prediction time.

A.5 Model Hyperparameter Settings

This appendix details the hyperparameter settings used for each machine learning model in the analysis. Table 12 summarizes the preprocessing methods, fixed parameters, and tuning grids for all models across different prediction horizons.

Table 12: Model Hyperparameter Specifications

Model	Horizon	Key Parameters	Search Grid
Panel A: Linear Models			
Common: Winsorization (5%, 95%), Standardization, max_iter=10			
Lasso	5s	-	$\alpha \in [3.0, 7.0] \times 10^{-6}$
(ℓ_1)	15s	-	$\alpha \in [0.3, 1.0] \times 10^{-5}$
	30s	-	$\alpha \in [0.9, 2.0] \times 10^{-5}$
	60s	-	$\alpha \in [2.0, 6.0] \times 10^{-5}$
Ridge	5s	-	$\alpha \in [1.0, 2.5] \times 10^{4}$
(ℓ_2)	15s	-	$\alpha \in [2.5, 6.0] \times 10^{4}$
	30s	-	$\alpha \in [0.5, 1.2] \times 10^{5}$
	60s	-	$\alpha \in [1.0, 2.5] \times 10^{5}$
Elastic Net	5s	-	$\alpha \in \{4, 6, 8\} \times 10^{-6}, \rho \in \{0.90, 0.95, 0.99\}$
($\ell_1 + \ell_2$)	15s	-	$\alpha \in \{1, 2, 3\} \times 10^{-5}, \rho \in \{0.70, 0.75, 0.80\}$
	30s	-	$\alpha \in \{6, 7, 8\} \times 10^{-5}, \rho \in \{0.25, 0.30, 0.35\}$
	60s	-	$\alpha \in \{4, 6, 8\} \times 10^{-4}, \rho \in \{0.05, 0.10, 0.15\}$
Panel B: Tree-Based Models			
Common: Winsorization (5%, 95%), Standardization			
LightGBM	5s	$N_{est} = 100$	Max Depth $\in \{2, \ldots, 8\}$
	15s	$N_{est} = 80$	Max Depth $\in \{2, \ldots, 8\}$
	30s	$N_{est} = 60$	Max Depth $\in \{2, \ldots, 8\}$
	60s	$N_{est} = 40$	Max Depth $\in \{2, \ldots, 8\}$
Random Forest	5s	$N_{est} = 40, \text{leaf} = 3e^{-4}$	Max Depth $\in \{8, \ldots, 18\}$
	15s	$N_{est} = 40, \text{leaf} = 3e^{-4}$	Max Depth $\in \{6, \ldots, 16\}$
	30s	$N_{est} = 40, \text{leaf} = 3e^{-3}$	Max Depth $\in \{4, \ldots, 14\}$
	60s	$N_{est} = 40, \text{leaf} = 3e^{-3}$	Max Depth $\in \{2, \ldots, 12\}$

Note: This table reports the hyperparameter configurations. N_{est} denotes the number of estimators. ρ denotes the l1_ratio for Elastic Net. For LightGBM, 'reg_alpha' is fixed at 0.5 and 'min_child_samples' at 6000. For Random Forest, 'max_features' is fixed at 0.1 and 'max_samples' at 0.2. All models use 'random_state=42'.

A.6 Variable Definitions and Summary Statistics

Table 13: Variable Definitions

Variable	**Definition**
	Panel A: Investor Composition and Market Frictions
Retail Trade Ratio	The daily ratio of trading volume originating from small orders (defined as orders below RMB 200,000) to total trading volume. This variable serves as a proxy for retail trading intensity and noise trading.
Net Institutional Flow	The daily net buying volume (buy volume minus sell volume) from large orders (defined as orders above RMB 200,000), normalized by total trading volume. Positive values indicate net institutional accumulation.
Margin Trading Target	A dummy variable that equals 1 if the stock is included in the list of eligible securities for margin trading and short selling ("Two-Rong" target list) on a given day, and 0 otherwise. Inclusion in this list significantly relaxes limits to arbitrage.
Northbound Holdings	The proportion of the stock's float shares held by foreign investors via the Northbound Stock Connect program. This variable proxies for the presence of sophisticated foreign institutional capital.
Margin Buy Balance	The ratio of the outstanding balance of margin purchases (leveraged buying) to the float market capitalization. This variable proxies for leveraged retail sentiment.
	Panel B: Stock Characteristics and Controls
Nominal Price (log)	The natural logarithm of the daily closing price.
Daily Return	The daily logarithmic return calculated from close-to-close prices.
Trading Volume (log)	The natural logarithm of the total daily trading volume in RMB.
Float Market Cap (log)	The natural logarithm of the daily free-float market capitalization.
Proportional Spread	The daily time-weighted average of the bid-ask spread relative to the mid-price, calculated from high-frequency snapshots.
Volatility	The realized volatility, calculated as the standard deviation of 15-second intraday returns.
Beta (vs. CSI 300)	The slope coefficient from the regression of the stock's intraday returns on the CSI 300 index returns.
R^2 (vs. CSI 300)	The coefficient of determination from the regression of the stock's intraday returns on the CSI 300 index returns, measuring systematic risk exposure.
Idiosyncratic Volatility	The standard deviation of residuals from the regression of the stock's intraday returns on the CSI 300 index returns.
Opening Return	The return from the previous trading day's closing price to the current trading day's opening price. This variable captures the overnight information shock and opening price gap.

Table 14: Summary Statistics of Key Variables

Variable	N	Mean	SD	Min	P25	Median	P75	Max
Panel A: Investor Composition and Frictions								
Retail Trade Ratio (%)	2,805,711	147.487	39.781	0.000	139.085	155.897	170.016	202.316
Net Institutional Flow (%)	2,805,711	-2.574	9.135	-100.012	-7.770	-2.018	2.318	100.751
Margin Trading Target (%)	2,805,711	0.500	0.500	0.000	0.000	0.000	1.000	1.000
Northbound Holdings (%)	2,805,711	0.759	2.030	0.000	0.000	0.000	0.560	35.140
Margin Buy Balance (%)	2,805,711	2.058	2.793	0.000	0.000	0.081	3.747	44.632
Panel B: Stock Characteristics								
Nominal Share Price (log)	2,805,711	2.580	0.995	-1.470	1.855	2.494	3.198	7.864
Daily Return (log)	2,805,711	0.000	0.033	-2.224	-0.015	0.000	0.013	3.017
Traded Volume (log)	2,805,711	$-\infty$	NaN	$-\infty$	9.971	10.984	11.967	17.533
Total Market Cap (log)	2,805,683	22.294	1.269	16.725	21.447	22.133	23.023	28.815
Proportional Spread	184,914	-0.018	0.211	-2.000	0.001	0.001	0.002	2.000
Volatility	184,914	0.001	0.001	0.000	0.001	0.001	0.002	0.018
Beta with CSI300	184,914	0.621	0.545	-39.103	0.264	0.559	0.918	15.638
R2 with CSI300	184,357	0.074	0.086	0.000	0.011	0.042	0.107	0.992
Idiosyncratic Volatility	184,357	0.001	0.001	0.000	0.001	0.001	0.001	0.017
Jump	2,805,711	-0.000	0.033	-0.856	-0.005	0.000	0.003	10.524
Panel C: Model Prediction R2								
RandomForest								
return 5s r2	171,860	0.139	3.769	-1410.678	0.090	0.144	0.208	1.000
return 15s r2	171,060	0.116	8.171	-3338.513	0.069	0.115	0.203	1.000
return 30s r2	172,060	0.102	3.241	-1268.213	0.037	0.077	0.173	1.000
return 60s r2	171,780	0.062	2.631	-828.300	0.003	0.035	0.126	1.000
Ridge								
return 5s r2	171,980	0.108	8.785	-3083.789	0.078	0.130	0.196	1.000
return 15s r2	171,980	0.026	28.127	-10010.115	0.061	0.104	0.192	1.000
return 30s r2	171,980	-0.014	32.714	-10720.185	0.036	0.073	0.164	1.000
return 60s r2	171,980	-0.082	44.591	-15794.521	0.010	0.039	0.115	1.000

Note: This table reports summary statistics. The sample comprises CSI 300 constituents. N denotes the number of observations. 'Margin Trading Target' is a dummy variable indicating eligibility for margin trading and short selling.

A.7 Sector-Level Analysis

Table 15: Predictability by Industry Sector

	5s Return R^2 (%)		5s Direction Accuracy (%)	
Sector	**Mean**	**Median**	**Mean**	**Median**
Technology	16.2	13.8	61.5	60.1
Financial	14.9	12.4	60.3	58.9
Consumer	15.1	12.7	60.6	59.2
Industrial	14.7	12.2	60.1	58.7
Energy	13.8	11.5	59.4	58.0
Materials	14.3	11.9	59.8	58.4
Healthcare	15.6	13.2	61.0	59.6
Utilities	13.2	10.9	58.9	57.5

Note: This table shows predictability patterns across different industry sectors using GICS sector classifications.

Figure 10: Time-Series Evolution of Predictability and Market Conditions

Note: Time-series plot of the HS300 index (blue), market volatility (red), and model predictability (R2). The figure illustrates that while market conditions fluctuate significantly (e.g., high volatility periods), the model's predictive power remains positive, though it varies inversely with extreme market turbulence.

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